

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

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paper doi:10.5281/zenodo.20706893 · code github.com/Vortex-AI/emem (Apache-2.0) · live emem.dev · method: whitepaper v3, emem @18adb67



THE PROBLEM

Agents that work together drift apart.

Each agent reads the world through its own searches and summaries. After a few handoffs two agents no longer agree on where a place is, what was measured or when, and neither can show the other which number is right. Agreement does not rescue them: after a shared summary, two models agreed only on wrong answers (R1).

THE WORLD WRITES



a 2.02 GB scene; the square marks the cell of the NDVI record (drawn 100 m wide)

OTHER WRITERS

Copernicus DEM · Overture Maps · weather · TESSERA (retired)
device gate: ships, no hardware enrolled yet

114

MCP tools, with an 18-tool core loop at emem.dev/ mcp; no key

168

published algorithms: flood, heat, crop, solar and more

118

wired measurements from 46 declared source schemes

2.57 M

signed entries in the append-only log, 30 Sep 2026

177

REST paths under /v1, all in the live OpenAPI document

10

client paths return one fact_cid: REST, MCP, A2A, SDKs, LangChain, LlamaIndex, stock blake3

Six contributions

each a mechanism in emem's code at 18adb67, each with a measurement on the live system

C1 An address for every observation

A value is keyed by where, what and when, and named by the hash of its exact bytes. The key finds a record; the name proves which one.

```
k = (cell64, band, tslot)
fact_cid = base32(BLAKE3-256(CBOR(f)))
token = emem:fact:<cell>:<fact_cid>
```

A **1,115-byte** record read from a **2.02 GB** scene travels as **84** characters **\$4**. Ten client paths, three runs each, re-derive one fact_cid from the served bytes.

C4 One token algebra, pixel to archive

The same naming covers a value, a field, a field through time, a set of fields, any file hashed where it lies, and one handle for many facts. Each family declares how strongly it binds.

```
fact → raster → cube → rasterset
tree = Merkle(chunks of a file, in place)
bundle = one handle over ≤ 256 fact_cids
```

A 443 × 453 px B04 field under one token **\$5**; a 5-scene cube whose members name the scene actually used **\$6**; a **238.7 MB** COG hashed in 292 chunks without moving it **\$2**; 8 records behind one 38-character line **\$16**.

C2 A memory on two clocks

Valid time (when the Earth was seen) and record time (when it was signed) are both in the hashed bytes. New evidence appends; nothing is rewritten.

```
as of t: argmax_signed_at ≤ t (signed_at, cid)
receipt binds (as_of_tslot, as_of_signed_at)
leaf = BLAKE3(0x00 || BLAKE3(CBOR(attestation)))
RFC 6962-shaped log over every attestation
```

Bengaluru elevation moved **918.0 → 915.07 m** when the upstream DEM changed **\$12** **\$13**. The May token still verifies; "as known on 15 Jun" still returns 918.0; the contradiction detector flags the pair as a provider substitution.

C5 Verification that needs no trust

A receiver checks bytes, place, arithmetic, source pixel, signature and log with stock libraries. Each record shows how deep it can be checked.

```
accept ⇔ BLAKE3(b) = cid ∧ cell(b) = cell(token) ∧ NDVI(DN(b)) = value ∧ COG[row, col] = DN(b) ∧ Ed25519(σ) ∧ leaf ∈ STH
depth: recomputable > rerunnable > trace-verified > checkpoint-signed > attester-only
```

Our script: **15 links, 17 checks, 17.9 s**, no emem code **\$22**. An agent with no tie to us wrote a clean-room verifier from /v1/verifier_spec: **725** requests, zero errors, five tampered receipts rejected, eight real defects found and since fixed.

C3 Machines write facts; agents write claims

Only emem's readers of registered archives, enrolled devices and operator-listed keys write the fact plane. Agents sign notes under their own keys: data, never facts, never instructions to the reader. When nothing qualifies, emem signs an absence.

```
cell(bytes) ≠ cell(token) → refused (HTTP 409)
absence = (cell, band, tslot, reason_cid, sources)
```

Venice: *no carriageway within 50 m* in Overture 2026-09-23.1, naming the public Parquet row groups it read (exhq61ps...). /v1/p1ane/ conformance tests live that no tool accepts a caller's value.

C6 Computation others can recompute

An agent derives a value over signed facts and emem re-runs the pure operation before recording it. A guard refuses a sentence whose number disagrees with the fact it cites.

```
derive accepted ⇔ ulp_gap(v_caller, v_emem) = 0 (≤ 4 for n-ary sum, mean)
guard: stated ≠ signed at the written precision → deny
```

Our NDVI change of **-0.1298** (25 Sep 2025 → 2026) was recomputed bit-identical and signed under our key **\$11**. The guard refuses "35 °C" citing a signed 28.0 °C **\$14**.

What agents do with it

eight jobs served today by the same tools; one memory for every model

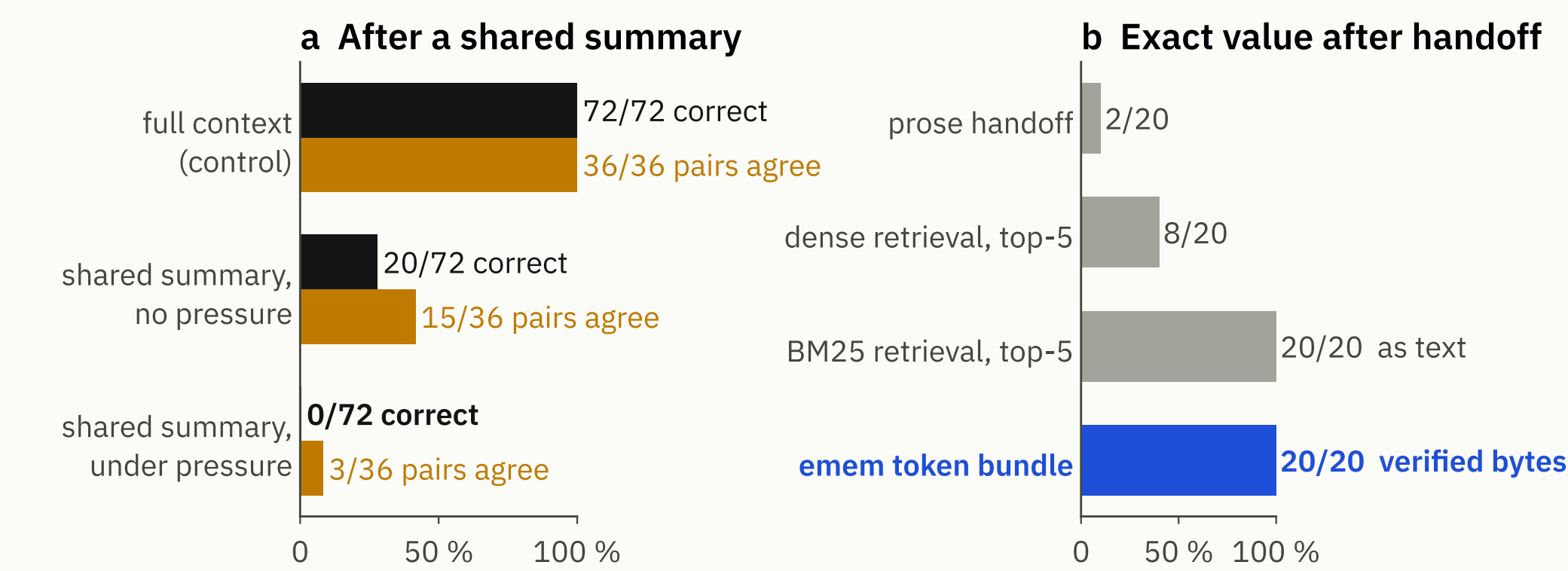
Research a place without the web ask · 168 recipes Flood, heat, crop or solar scores from signed readings; each recipe cited by id.	Hand work to another agent emem:bundle Exact signed bytes behind one line, on any model or vendor.	Keep a long investigation alive emem_memory_create Signed notes outlast sessions, compaction and restarts. Notes are public.	Agree on what a thing is emem_entity_resolve A farm or a building gets one identity; different phrasings converge on it.	Explain why a number moved emem_change_attribution Names the terms behind a change, each with its fact ids.	Catch a contradiction emem_memory_contradictions Finds records that disagree; the guard refuses a wrong number.	Turn documents into evidence emem:tree Lab reports and land records become signed units under one token.	Compute so others can recompute emem_derive Pure operations are re-run by emem before the result is recorded.
One token, three vendors Claude (through MCP), Gemma 3 on Amazon Bedrock and Qwen 2.5 running locally all end with the same fact_cid and value. The decoding is emem's, not the model's: two of the three were handed the same resolved response.		A public dispute, settled by bytes geo.qa's agent re-derived a Doha road fact and reported 9.8 m against emem's 5.4 m. Re-measuring from the full-precision coordinate inside the fact gave 5.4 m, and that agent withdrew its number in a signed note.		Regulation an auditor can re-check eudr.dev checks farm plots against the EU Deforestation Regulation cut-off with emem's forest facts, and prepares Annex II statements that an auditor can re-verify.		Any node, the same receipts The hosted node runs the Apache-2.0 binary, and a receipt minted on one node verifies on another. geo.qa runs a second node; an air-gapped build runs with no network at all.	

Four results

pre-registration hashes, data and scripts: github.com/Vortex-AI/esa_poster/research/repro

R1 · COMPACTION AND HANDOFF

Agreement is not evidence; bytes are



(a) Pre-registered; Gemma-4-12B and Qwen2.5-7B on one host **\$17**. Under pressure 0 of 72 answers were right, yet 3 of 36 pairs agreed, every time on a wrong answer (registered one-sided Fisher test, $p = 0.035$; pairs and answers share trials, so read it as descriptive). (b) $n = 20$ per arm **\$18**; a single-token arm was excluded for a window bug. BM25 ties on exactness; only the bundle hands over bytes the receiver can re-hash.

| Agreement between agents cannot certify a value. A content address can.

R2 · FALSIFIABLE BY CONSTRUCTION

The records located emem's own error



• query point the pixel that contains it the pixel whose DNs emem signed

Re-reading the scenes named inside the records showed that before 28 Sep, **162 of 200** sampled Sentinel-2 records (81 %, Wilson 95 % CI 75–86 %) carried the DNs of a neighbouring pixel; here, 10 m south **\$7**, emem rounded a fractional pixel index where GDAL takes the floor. After the fix **\$8**, 49 of 49 records where the two rules differ read the containing pixel. No signed byte was rewritten, so every affected record stays resolvable and datable.

| Any memory will hold errors. In emem each one can be found, dated and bounded.

R3 · TWO LLMS, EMEM THE ONLY TOOL

The check belongs in the tool, not the model

B RECEIVES	B RIGHT	B CHECKED
A's token	10/10	resolved; harness re-hash, receipt 10/10
A's prose, 16 digits	10/10	nothing to check
prose rounded to "0.47"	0/5	nothing to check
token with a forged cell	5/5 declined	MCP error; REST 409

A = Claude Sonnet 5.5, B = Claude Haiku 4.5, pre-registered **\$22**. Token vs full prose tied ($p = 1$); the rounded arm is exploratory, its threshold set 0.0004 below the value. A registered small-model arm (Qwen2.5-3B as B) did worse; its raw logs are not yet committed, so its numbers are not printed.

| A token protects an agent only when something checks it. The check belongs in the resolver, not in the model's goodwill.

R4 · SEVEN REAL DRIFTS

Six are caught by one test on the record

WHAT DRIFTED, REAL CASE	TEST THAT SHOWS IT
words: 0.48715... retold as "≈ 0.49" \$21	s ≠ v
reader: a neighbouring pixel \$7	DN(b) ≠ COG[row, col]
source: DEM 918.0 → 915.07 m \$13	fn_key₁ ≠ fn_key₂
date: cube member for 15 May is 10 Jun \$6	date(scene) ≠ asked
place: token relabelled to a cell \$4	cell(b) ≠ cell(token)
signer: one value, 7 fact_cids \$13	k₁ = k₂, cid₁ ≠ cid₂
referent: "Maasvlakte ramp 7"	none: a label

Objections we expect

and our answers

A signature does not make a value true.

AGREED It proves who served which bytes, and when. Truth is checked by re-reading the public source each record names. That is how R2 was found.

A token costs more LLM tokens than the number it names.

AGREED About 6× for one value in our count (46 vs 8 tokenizer tokens). A single token buys identity, not compression. The bundle compresses: up to 256 facts behind one 38-character line.

Why not STAC with checksums, C2PA, W3C PROV or IPFS?

REBUTTAL They locate files, sign media, describe lineage or address blobs. None names one value at a place and time, binds it to its cell and answers "as of t". emem works at the grain agents cite.

One operator runs the responder and the log.

AGREED 111 keys co-sign the log, but the only domain-vouched one, geo.qa, is also Vortex AI. Every check above runs without us, and an unaffiliated clean-room verifier reproduced them (C5). Independent operators are the next step.

The title says foundation-model embeddings, yet the encoders were retired.

REFRAME Clay, Prithvi-EO-2.0, Galileo and a JEPA head left the code after acceptance **\$23**, and emem.dev now mints no new TESSERA vectors. Vectors signed before still resolve and verify **\$24**: the memory outlived its encoders.

BM25 retrieval matched you.

REBUTTAL On exactness, when the string is in the corpus (16/16). Retrieval returns text nobody can check; dense retrieval gave 4/142 exact and up to 138 confidently wrong, by a median 252 m **\$25**.

Your own reader signed wrong-pixel records.

AGREED 162 of 200 before 28 Sep. The records made the error locatable by re-reading the scene they name, and datable by their signed_at. A store of bare vectors could show only that a value moved.

Small n, one host.

AGREED The agent studies are samples: pre-registered, hashed before trial 1, not yet replicated. The mechanism claims (hashing, cell binding, signatures) are deterministic and need no sample.

THE INVENTION

One place, one address. One observation, one signed record. The record's name, not a paraphrase, is what crosses between agents.

emem is a memory of the physical world that agents share and no agent controls. It is **written by machines** from public archives, **named by the hash** of its bytes and **kept on two clocks**. Anyone can check it: bytes, signature and log offline, and the source pixel with one public read.